Reproducing and Extending Stochastic Switching in a Probabilistic Bandit Task Using Stickiness-Based Decision Models

Gillian Macdonald, Lenka Okasova, Bob Rice, and Ningqian Yang*
School of Mathematical Sciences, University of Nottingham, UK

Adaptive decision-making in uncertain environments requires integrating reward history to guide flexible action selection. Beron et al. (2022) [1] showed that mice performing a probabilistic two-choice bandit task exhibit stochastic, history-dependent switching behaviour that is accurately captured by reinforcement learning and recursive logistic regression models incorporating choice history bias ("stickiness"), whereas inference-based hidden Markov models (HMMs) exhibit excessive switching. Here, we reproduce the key behavioural and generative modelling results under the original 80–20 reward condition and extend the analysis to a noisier 70–30 environment. We compare the behaviour of the recursive logistic regression (RFLR) model and HMM using reward-based performance metrics, including reward rate, switching frequency, and reward-per-switch efficiency, and examine how behavioural parameters adapt across environments. We introduce nonlinear stickiness mechanisms by implementing temporally gated and prediction-error–dependent modulation of choice history bias and assess whether adaptive stickiness improves switching efficiency and behavioural stability. Model performance is evaluated using generative simulations and behavioural efficiency metrics. Our results demonstrate that stickiness increases under environmental uncertainty and that adaptive stickiness mechanisms improve behavioural efficiency by reducing excessive switching while preserving reward. These findings suggest that stickiness functions as a dynamically regulated control variable that stabilises behaviour and supports efficient decision-making under uncertainty.

INTRODUCTION

Adaptive decision-making in uncertain environments requires balancing behavioural flexibility with persistence. In probabilistic reversal learning tasks, agents must integrate reward history to infer changes in environmental contingencies while avoiding overreaction to stochastic outcomes. A central feature of such behaviour is stochastic switching, in which agents occasionally switch actions despite recent reward, reflecting uncertainty about the underlying state. In rodents and humans, probabilistic reversal learning engages cortico-striatal circuits implicated in reinforcement learning and behavioural flexibility, including orbitofrontal cortex, medial prefrontal cortex, and dorsal striatum [2, 3]. Dopaminergic signalling is thought to encode reward prediction errors that guide value updating, while prefrontal regions contribute to monitoring volatility and regulating behavioural persistence. Disruptions to these systems alter switching behaviour, producing either excessive perseveration or oversensitivity to negative feedback. Thus, understanding how agents regulate persistence under stochastic reward conditions has both computational and biological significance. Beron et al. (2022) [1] addressed this problem in mice performing a dynamic two-choice probabilistic bandit task. They showed that mouse behaviour deviates from that of an ideal Bayesian observer performing hidden Markov model (HMM) inference, particularly in its switching dynamics. Instead, behaviour was accurately captured by a recursive logistic regression (RFLR) model incorporating a choice-history bias ("stickiness") and a stochastic action policy. Importantly, these models achieved near-maximal reward rates despite deviating from theoretically optimal inference, suggesting that behavioural persistence functions as a stabilising mechanism under stochastic feedback. Although Beron et al. demonstrated that stickiness is necessary to explain mouse behaviour, the computational role of this parameter remains underspecified. In their formulation, stickiness was treated as a fixed bias that varies across reward contexts but does not dynamically adapt within a session. It therefore remains unclear whether stickiness reflects a static behavioural tendency or a control variable that is regulated online in response to environmental volatility. If stickiness functions to buffer against stochastic reward fluctuations, one would predict that it should scale with environmental uncertainty and potentially respond to recent prediction error signals. Testing this hypothesis requires extending the fixed-stickiness framework to include adaptive modulation mechanisms. In this study, we first establish a behavioural and modelling baseline using the framework introduced by Beron et al. (2022) under the original 80–20 reward condition. We then extend the analysis to a noisier 70–30 environment to examine how behavioural parameters scale with increased uncertainty. Finally, we introduce nonlinear, adaptive stickiness mechanisms and evaluate their effects on switching behaviour and reward efficiency using generative simulations and cross-validated predictive analyses. By integrating a validated baseline with targeted model extensions, we aim to clarify the computational and potentially biological role of stickiness as a dynamically regulated mechanism for behavioural persistence under uncertainty.

METHODS: BASELINE FRAMEWORK FROM BERON ET AL. (2022)

Behavioural signature of ineterest in the original study

Beron et al. (2022) demonstrated that mice performing a dynamic two-choice probabilisticbandit task exhibit graded, stochastic switching rather than deterministic win–stay/lose–switch behaviour. The key behavioural signature is the conditional switching probability $P(\text{switch} \mid \text{history})$, which increases with accumulating negative evidence but remains bounded below one. Despite stochastic feedback, mice achieve near-maximal reward rates, indicating efficient adaptation under uncertainty.

Baseline implementation and data sources

To establish a validated behavioural and modelling baseline, we used the publicly available dataset and analysis/model code released by Beron et al. (2022). We executed the authors' pipeline to regenerate the key behavioural summaries and model-comparison figures reported in the original study. All baseline figures were generated using the provided scripts/notebooks with fixed random seeds where applicable, and outputs were cross-checked against the published results for qualitative agreement (e.g., graded policy structure, model ranking, and HMM overswitching).

Computational models considered in Beron et al. (2022)

Beron et al. (2022) compared multiple model classes that differ in how reward history and environmental uncertainty are represented. We summarise these model families here to define the computational hypotheses evaluated in both the reproduction and extension analyses.

Win–Stay/Lose–Switch (WSLS). A heuristic strategy that repeats rewarded actions and switches after unrewarded outcomes. WSLS predicts near-deterministic switching after losses and therefore fails to capture graded behavioural responses.

Reinforcement learning (RL). A model-free approach that incrementally updates action values via reward prediction errors, typically paired with stochastic action selection. RL captures gradual adaptation but does not explicitly represent latent states of the environment.

Inference-based models (HMM). A latent-state model that infers the current reward contingency from past outcomes using Bayesian belief updating. Inference models can adapt rapidly following reversals but may over-switch under stochastic feedback if stochastic reward omissions are over-interpreted as evidence for latent state change. Beliefs are updated according to:

$$P(s_t \mid o_{1:t}) \propto P(o_t \mid s_t) \sum_{s_{t-1}} P(s_t \mid s_{t-1}) P(s_{t-1} \mid o_{1:t-1}),$$

where $P(s_t \mid o_{1:t})$ is the posterior belief over latent reward states given observations $o_{1:t}$. For Sticky HMM variants, PE_t was defined as the difference between the observed reward and the belief-weighted expected reward under the model's current posterior belief and emission probabilities. Here, observations o_t correspond to trial outcomes (reward) conditioned on the chosen action at time t.

Recursive logistic regression (RFLR). A reduced model that predicts choices from accumulated choice–outcome history while incorporating a choice-history bias (stickiness). In the recursive formulation, the probability of choosing action a_t is:

$$P(a_t = 1) = \frac{1}{1 + \exp\left[-(\beta x_t + \alpha a_{t-1})\right]},$$

where x_t represents accumulated reward evidence, β is an inverse temperature controlling sensitivity to evidence, and α is the stickiness parameter biasing repetition of the previous choice. Additional analysis scripts were used to regenerate behavioural summary metrics reported in the original study, including conditional switching probabilities, event-triggered switching dynamics, and quantitative comparisons between model and mouse policies. These analyses confirmed qualitatively with the key behavioural signatures described by Beron et al. (2022) and established the modelling framework used for subsequent extensions.

METHODS: EXTENSIONS

Behaviour under increased uncertainty (70–30 condition)

Generative model simulations. Using cross-validated parameter estimates, we generated synthetic behaviour in both reward environments (80-20 and 70-30) under three model classes: RFLR, Ideal HMM (Thompson sampling), and Sticky HMM. To ensure identical reversal structure across models, simulations were run on the empirical latent state sequences extracted from the experimental sessions. Reward delivery probabilities were set to 0.8/0.2 (80–20) or 0.7/0.3 (70–30). Transition probabilities were set to the task transition parameter used in the generative simulations (p_{switch}), consistent

with the probabilistic bandit structure. For each environment, RFLR parameters (α, β, τ) estimated via cross-validation were used to simulate behaviour. To assess parameter generalisation across environments, we additionally simulated the 70-30 condition using parameters fitted in the 80-20 environment.

Cross-validated parameter estimation across reward environments. To quantify how behavioural parameters depend on environmental uncertainty, the RFLR model was fit separately in the 80–20 and 70–30 reward environments. Parameter estimation was performed using 5-fold cross-validation at the session level to ensure robust generalisation and prevent information leakage between training and evaluation data. Sessions were randomly partitioned into five folds using a fixed random seed. For each fold, model parameters (α, β, τ) were fit on four training folds by maximising the log-likelihood of the observed choice sequence under the RFLR model. Optimisation was implemented using stochastic gradient descent (10,000 iterations; learning rate = 0.01). The fitted parameters were then evaluated on the held-out fold to assess out-of-sample predictive performance. This procedure was repeated for all folds, and the final parameter estimate for each environment was taken as the mean of parameter estimates across folds.

Behavioural metrics and model comparison. Model behaviour was evaluated using summary statistics computed at the session level. For each simulated session, we computed the total reward obtained per session and the total number of switches per session. These metrics were used to characterise the reward-cognitive trade-off across models. To quantify switching dynamics around environmental reversals, we computed the probability of switching as a function of trial distance relative to block transitions. Switch probabilities were aligned to empirical state transitions and averaged across sessions to characterise cognitive lag and adaptation dynamics. Finally, parameter estimates (α, β, τ) were compared across the 80-20 and 70-30 environments to assess systematic shifts associated with increased reward uncertainty.

Nonlinear stickiness mechanisms

To extend the fixed-stickiness framework, we implemented dynamic modulation of the stickiness parameter (α) based on recent environmental feedback. These mechanisms were applied to the RFLR, HMM, and forgetting Q-learning (F-Q) architectures and evaluated using generative simulations.

Rolling window statistic Let W_t denote the set of indices corresponding to the most recent N trials up to time t (or fewer if $t < N$). We define the rolling statistic as

$$\bar{E}_t = \frac{1}{|W_t|} \sum_{k \in W_t} \mathrm{PE}_k. \tag{1}$$

This statistic was updated sequentially on each trial using a fixed-length rolling buffer.

Informational gating: signed prediction-error modulation (RFLR and HMM). For the RFLR and HMM models, stickiness was dynamically scaled as a function of recent signed prediction error (PE). On each trial $t > 0$, effective stickiness was computed via exponential gating:

$$\alpha_t = \alpha_{\text{base}} \exp\big(\gamma\, \bar{E}_t\big)\,. \tag{2}$$

where α_{base} is baseline stickiness, γ is surprise sensitivity, and $\bar{E}_t$ is the mean PE over an N-trial rolling window (defined above).

Prediction-error signals. Prediction-error signals were defined to match the implementation for each model class. For dynamic RFLR, we used an internal expected-reward proxy $\hat{r}_t = \sigma(\phi\, c_t)$ derived from the model's choice-weighted term c_t, and defined

$$\mathrm{PE}_t \;=\; r_t - \hat{r}_t, \qquad \mathrm{PE}_0 \;=\; r_0 - 0.5. \tag{3}$$

For Sticky HMM variants, PE_t was defined as the difference between the observed reward and the belief-weighted expected reward under the model's current posterior belief and emission probabilities. In the nonlinear Sticky HMM implementation, emission probabilities were parameterised asymmetrically with respect to chosen action, such that the reward likelihood under each latent state depended explicitly on the action taken at trial t. The belief-weighted expected reward was therefore computed using this action-conditioned emission structure, consistent with the computational implementation. For the F-Q nonlinear stickiness variant, the gating signal used the absolute value error

$$\varepsilon_t \;=\; \big|\, r_t - Q_t(a_t) \,\big|, \tag{4}$$

and the rolling window statistic was defined as

$$\bar{E}_t = \frac{1}{|W_t|} \sum_{k \in W_t} \varepsilon_k, \tag{5}$$

where W_t denotes the set of indices corresponding to the most recent N trials (or fewer if $t < N$). This rolling mean was updated sequentially at each trial and used to modulate stickiness via exponential gating.

Multi-level gating framework and optimal evaluation

Model hypotheses. Phase 4 evaluates distinct computational hypotheses for how agents use environmental

"surprise" to regulate choice persistence in the 2ABT. We compare (i) a direct policy model where surprise gates motor persistence trial-by-trial (dynamic RFLR), (ii) a belief-based model where Bayesian state inference is combined with motor-level stickiness (Sticky HMM), and (iii) a temporal/value baseline where switching behaviour emerges from windowed value dynamics (FQ value-gated). In addition, belief-reset variants test whether omission-driven surprise triggers change-point detection and belief reinitialisation.

Cross-validated optimisation. To prevent overfitting, parameters were fit using 5-fold cross-validation over sessions. For each fold, model parameters were optimised on the training set and evaluated on held-out sessions using mean log-likelihood (Mean LL) as the primary performance metric.

Nested search over temporal parameters. Discrete temporal parameters (e.g., commitment window N) were selected via a nested search procedure. The outer loop enumerated candidate N values, and the inner loop performed continuous optimisation of model parameters (e.g., α_{base} and surprise sensitivity γ) using L-BFGS-B with bounds. Parameter stability was assessed via across-fold variability, and the optimal N was selected via plurality vote across folds.

Model design: temporal gating and surprise-dependent modulation. To extend the computational framework beyond fixed stickiness, we implemented nonlinear stickiness mechanisms that dynamically regulate behavioural persistence. These extensions were applied to RFLR, HMM, and forgetting Q-learning (F-Q) models. Temporal gating was implemented at the value level within the F-Q architecture. Q-values were held fixed during each N-trial window and updated only at window boundaries using the mean reward obtained for each action during that window. The agent continued to make a free stochastic choice on every trial based on the current (potentially frozen) Q-values, so temporal gating discretised the learning signal rather than the motor output. Informational gating was implemented via prediction-error–dependent modulation of stickiness. For RFLR and HMM variants, the gating signal used *signed* prediction error PE_t. For the F-Q nonlinear stickiness variant, the gating signal used *absolute* value error $\varepsilon_t = |r_t - Q_t(a_t)|$.

Simulation protocol and recorded outputs. Synthetic behavioural sessions were generated using the same probabilistic bandit task structure described above. Models were simulated across multiple sessions, and we recorded the dynamic stickiness trace, switching frequency, and (where applicable) value estimates (e.g., Q-values for F-Q). These outputs were used to assess how adaptive stickiness regulates the balance between persistence and flexibility under stochastic feedback. **Implementation details.** All extension simulations were implemented in Python using custom code written for this project. For nonlinear stickiness experiments, synthetic behavioural data were generated for 5 independent sessions of 120 trials each, with reward contingencies evolving according to a two-state Markov process with transition probability $P(\text{switch}) = 0.1$. Behavioural metrics were averaged across sessions. Dynamic stickiness was modulated according to the mean prediction error over an N-step rolling window, with parameter values matched to those used in the computational implementation. Temporal gating was implemented by discretising Q-value updates into N-trial windows, with Q-values held fixed within each window and updated using the mean reward per action at window boundaries, while informational gating was implemented by updating the stickiness parameter on each trial according to the accumulated prediction error window. We hypothesised that dynamic stickiness would improve switching efficiency by preventing excessive switching in response to stochastic reward fluctuations and, where evaluated against baseline models, improve alignment with mouse switching dynamics. **Reproduction: quantitative agreement for all models.** Model-predicted vs mouse-observed $P(\text{switch} \mid \text{history})$ across histories (dotted identity line).

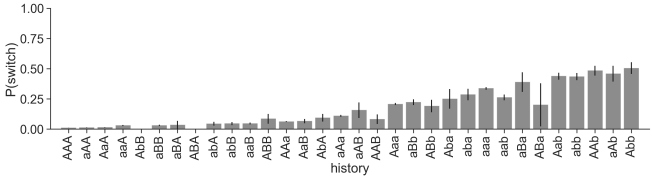


FIG. 1: **Mouse baseline exhibits graded, stochastic switching.** Conditional switching probability $P(\text{switch} \mid \text{history})$ computed nonparametrically from mouse behaviour (history length = 3). Switching probability increases with accumulating negative evidence but remains bounded below one, demonstrating graded, stochastic rather than deterministic win–stay/lose–switch behaviour.

RESULTS

Reproduction of Beron et al. (2022)

We first verified that our implementation reproduces the core behavioural signatures reported by Beron et al. [1]. Conditional switching probabilities computed directly from mouse behaviour exhibited graded dependence on recent reward history and remained bounded below one (Fig. 1), confirming stochastic, history-dependent switching.

We next examined behaviour around contingency reversals. Mice exhibited gradual adaptation in both choice probability and switching rate. Logistic regres-

sion, RFLR, and forgetting Q-learning reproduced this gradual adjustment, whereas the HMM showed sharper transitions and elevated switching (Fig. 2).

Differences were also evident in conditional switching structure (Fig. 3). The RFLR closely matched the graded switching policy observed in mice, whereas the HMM systematically overestimated switching probability across many histories.

Scatter comparisons (Fig. 4) confirmed that RFLR provided the closest quantitative match to mouse switching behaviour, whereas HMM deviated substantially.

Together, these results reproduce the main findings of Beron et al., demonstrating that mouse behaviour is best captured by models incorporating stochastic evidence integration and choice history bias.

Extension: Behaviour under increased uncertainty (70–30)

We next examined behaviour in a noisier 70–30 reward environment.

Cross-validated parameter estimates revealed systematic modulation across environments (Fig. 5). Stickiness increased in the 70–30 condition, consistent with greater behavioural persistence under increased reward stochasticity.

Behavioural efficiency revealed a clear reward–switch trade-off (Fig. 6). Although the HMM achieved slightly higher reward, it did so at the cost of substantially increased switching.

Reversal-aligned switching dynamics further demonstrated environment-specific effects (Fig. 7). Parameters fitted in the 70–30 condition reproduced gradual adaptation, whereas applying 80–20 parameters led to exaggerated switching.

Finally, plotting reward-per-switch against switching frequency (Fig. 8) highlighted differences in behavioural efficiency. The HMM occupied a high-switch, low-efficiency regime, whereas RFLR remained closer to empirical mouse behaviour.

Extension: Adaptive stickiness mechanisms (Generative evaluation)

To determine whether dynamic regulation of stickiness improves behavioural efficiency, we evaluated generative simulations of adaptive variants across reward environments.

We first compared static and dynamic RFLR models (Fig. 9). Dynamic stickiness shifted behaviour toward lower switching rates while preserving reward, particularly under the noisier 70–30 condition. This indicates that adaptive modulation of persistence improves the reward–cost trade-off relative to fixed parameters.

We next compared all adaptive variants jointly (Fig. 10). Across both reward environments, dynamic RFLR, sticky HMM variants, and value-gated FQ models occupied regions of higher reward-per-switch efficiency than inference-only baselines. Notably, symmetric and asymmetric reset mechanisms reduced excessive switching in the HMM without sacrificing overall reward.

Finally, reversal-aligned generative transition dynamics revealed how adaptive mechanisms stabilise behaviour across environments (Fig. 11). Under the 80–20 condition, all models adapt rapidly following reversals. In contrast, under 70–30 uncertainty, adaptive stickiness produces more gradual, bounded switching dynamics, preventing oversensitive responses to stochastic losses.

Together, these generative results demonstrate that dynamic regulation of stickiness improves behavioural efficiency and produces switching dynamics that more closely resemble empirical mouse behaviour, particularly under increased reward stochasticity.

Predictive validation of adaptive stickiness mechanisms

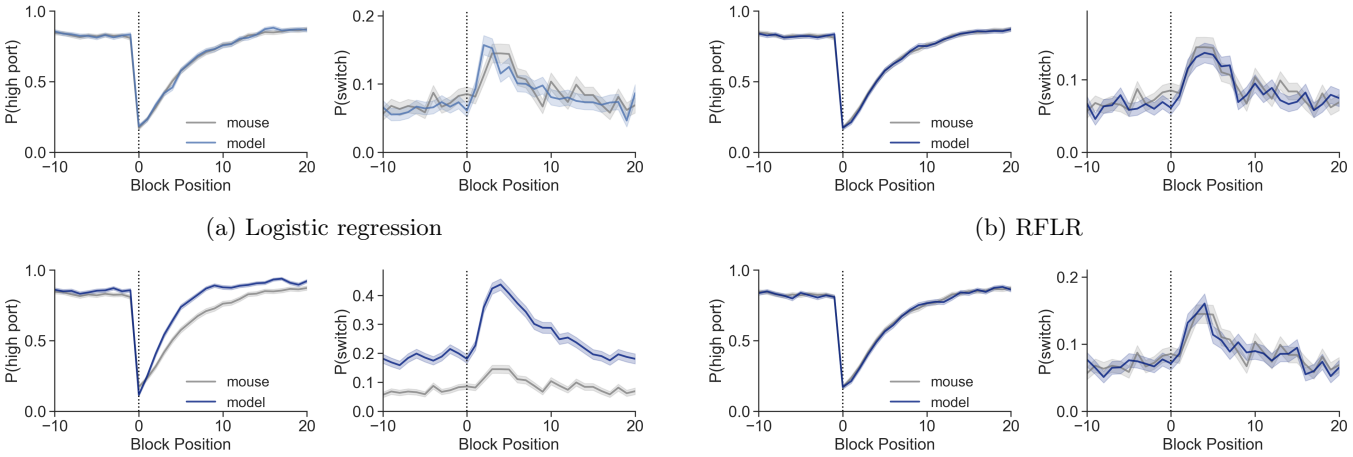
We next evaluated whether adaptive stickiness mechanisms capture the fine-grained structure of mouse switching behaviour beyond reward-based efficiency metrics.

History-dependent switching structure. Figure 12 shows conditional switching probabilities across all 32 three-trial reward histories for the dynamic RFLR model under both reward environments. Under 70–30 uncertainty, dynamic RFLR reproduces the graded structure of mouse switching probabilities, maintaining bounded switching even after streaks of losses. Under 80–20, switching remains sharply modulated by recent reward history, consistent with empirical behaviour.

Sticky HMM with symmetric reset exhibits similar graded structure (Fig. 13), but produces slightly elevated switching following rare loss streaks, particularly under 70–30 uncertainty.

Quantitative policy agreement. We quantified model–mouse agreement by correlating predicted and empirical conditional switching probabilities across histories (Fig. 14). Dynamic RFLR exhibits strong alignment across both reward environments, whereas Sticky HMM shows moderate but improved agreement relative to inference-only variants. These results confirm that adaptive stickiness mechanisms capture the detailed shape of mouse switching policy rather than merely reproducing aggregate reward statistics.

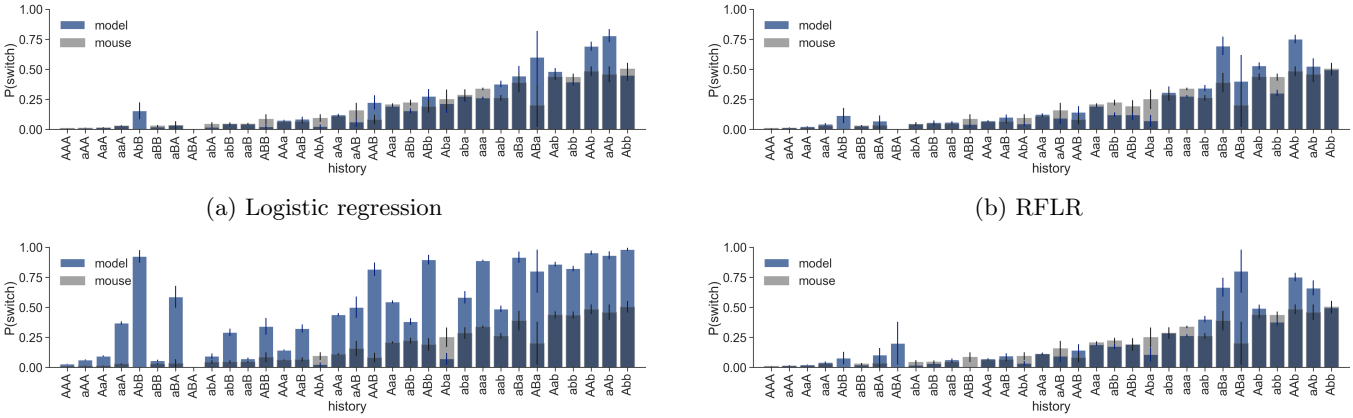
Reversal-aligned predictive dynamics. Finally, we evaluated predictive reversal dynamics across environments. As shown in Fig. 15, dynamic RFLR closely tracks mouse adaptation following contingency reversals under both 80–20 and 70–30 conditions, maintaining bounded switching and avoiding oversensitivity to



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 2: **Model comparison around contingency reversals.** Choice probability (left) and switching probability (right) aligned to block transitions. RFLR and logistic regression reproduce the gradual adaptation observed in mice, whereas the HMM exhibits sharper transitions and elevated switching, reflecting oversensitivity to stochastic reward omissions.



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 3: **Conditional switching structure reveals HMM overswitching.** Mouse (grey) and model (blue) conditional switching probabilities across recent reward histories. RFLR closely matches the graded switching structure of mouse behaviour, while the HMM systematically overestimates switching across multiple histories.

stochastic outcomes. Sticky HMM with symmetric reset similarly stabilises reversal-aligned switching relative to inference-only models, though with slightly steeper transitions.

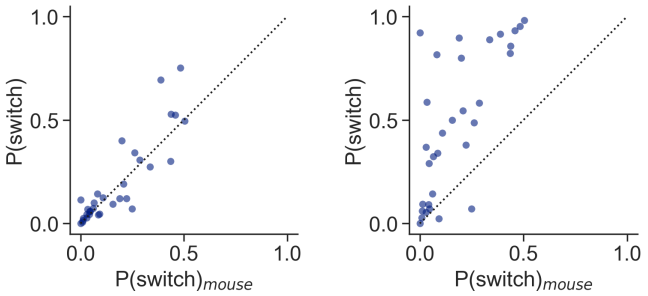
Collectively, these predictive analyses demonstrate that adaptive stickiness mechanisms not only improve reward-per-switch efficiency but also reproduce the structured, graded, and environment-sensitive switching policy observed in mice.

Cross-validated predictive comparison of adaptive stickiness models

To complement reward–switch efficiency analyses, we compared adaptive model variants using held-out predictive log-likelihood from 5-fold cross-validation (Table I). Across both reward environments, dynamic RFLR achieved the highest predictive performance (least negative Mean LL; Table I). In the 80–20 condition, dynamic RFLR improved held-out log-likelihood by approximately 0.0028 per trial relative to the next-best adaptive variant (HMM resetv2), and by 0.0056 per trial in the 70–30 condition. Although modest at the single-trial level, these differences accumulate across sessions

TABLE I: **Cross-validated predictive performance and fitted adaptive-stickiness parameters.** Mean held-out log-likelihood (Mean LL), fitted base stickiness α_{base} (mean ± SD across folds), surprise sensitivity γ (mean ± SD), and selected commitment window N for each model and reward environment. Higher Mean LL (less negative) indicates better predictive performance on held-out data.

Condition	Model	Mean LL	α_{base}	SD	γ	SD	N
80–20	RFLR	-0.1804	1.052	0.010	0.680	0.020	1
80–20	HMM_sticky	-0.2318	1.227	0.020	0.460	0.020	1
80–20	HMM_reset	-0.1984	0.606	0.010	0.883	0.010	1
80–20	HMM_resetv2	-0.1832	0.809	0.010	0.661	0.010	1
80–20	FQ	-0.2204	2.076	0.010	0.002	0.000	2
80–20	FQ_Value_Gated	-0.2204	2.077	0.010	0.000	0.000	2
70–30	RFLR	-0.1959	1.352	0.020	0.506	0.010	1
70–30	HMM_sticky	-0.2447	1.392	0.020	0.375	0.010	1
70–30	HMM_reset	-0.2094	0.818	0.010	0.854	0.010	1
70–30	HMM_resetv2	-0.2015	0.997	0.010	0.561	0.010	1
70–30	FQ	-0.2356	1.702	0.010	0.548	0.020	2
70–30	FQ_Value_Gated	-0.2278	2.172	0.010	0.000	0.000	2



(a) RFLR (b) HMM

FIG. 4: **Quantitative model–mouse agreement.** Scatter plots comparing model-predicted and empirical $P(\text{switch} \mid \text{history})$ values. Points closer to the identity line indicate better agreement. RFLR demonstrates stronger quantitative alignment with mouse policy than the HMM.

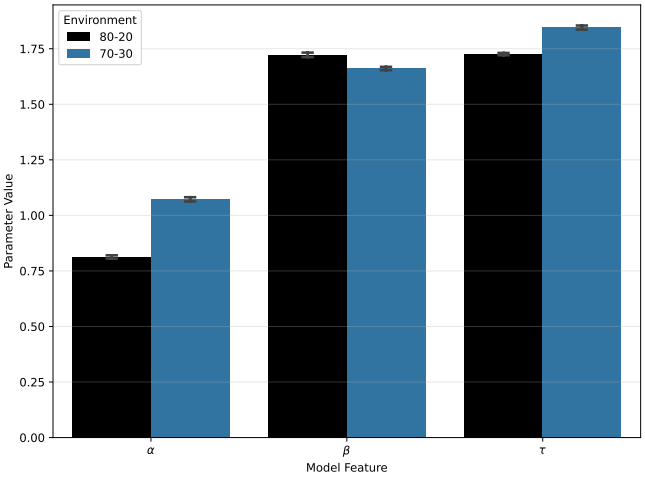


FIG. 5: **Stickiness increases under environmental uncertainty.** Cross-validated RFLR parameter estimates across 80–20 and 70–30 reward environments. Stickiness (α) increases systematically in the noisier 70–30 condition, indicating greater behavioural persistence under increased reward stochasticity.

and indicate consistently superior out-of-sample predictive performance. Although adaptive variants introduce additional parameters (e.g., γ and commitment window size N), improvements in held-out likelihood indicate that gains reflect improved predictive structure rather than mere parameter flexibility. Parameter estimates also showed systematic context-dependence: in dynamic RFLR, base stickiness increased from $\alpha_{\text{base}} = 1.052 \pm 0.010$ in 80–20 to $\alpha_{\text{base}} = 1.352 \pm 0.020$ in 70–30, consistent with greater behavioural persistence under increased reward stochasticity. In contrast, surprise sensitivity decreased from $\gamma = 0.680 \pm 0.020$ (80–20) to $\gamma = 0.506 \pm 0.010$ (70–30), suggesting reduced trial-by-trial reactivity under higher uncertainty. Across models, the selected commitment window was typically short ($N = 1$), with value-gated FQ favouring a slightly longer window ($N = 2$), consistent with discretised value updating. Together, these results provide likelihood-based support for adaptive stickiness mechanisms and quantify how fitted persistence parameters shift with environmental uncertainty.

Normative interpretation: efficiency versus optimal inference

To contextualise adaptive stickiness mechanisms, we compared their behaviour to an inference-based benchmark (Ideal HMM with Thompson sampling), which approximates normative Bayesian state inference in the

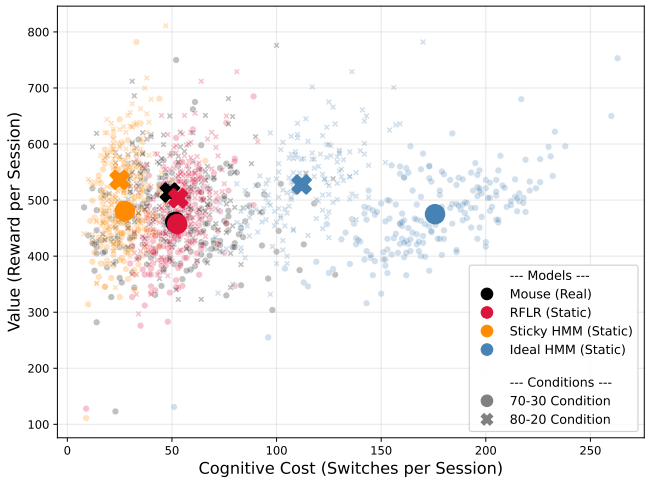


FIG. 6: **Reward–switch trade-off across environments.** Total reward per session plotted against switching frequency. While inference-based HMM achieves slightly higher reward, it does so at the cost of substantially increased switching, resulting in lower reward-per-switch efficiency compared to RFLR and mouse behaviour.

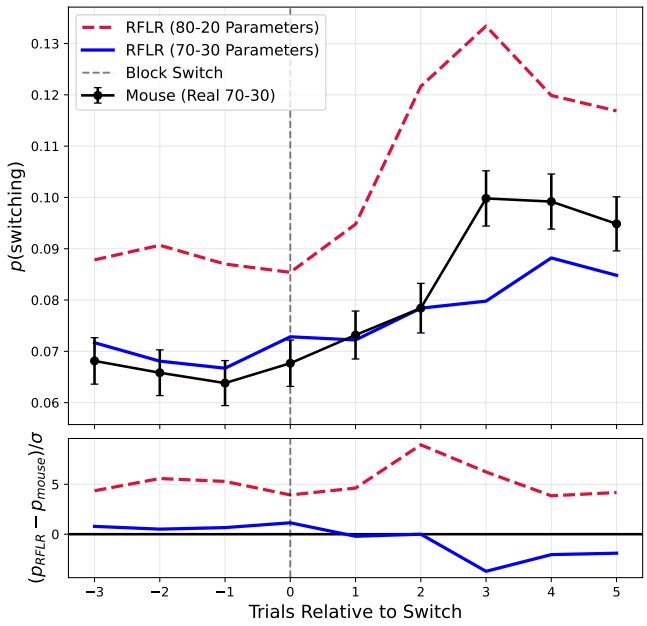


FIG. 7: **Cross-environment parameter transfer induces oversensitive switching.** Reversal-aligned switching dynamics under the 70–30 condition using parameters fit in either 80–20 or 70–30 environments. Applying 80–20 parameters exaggerates switching following reversals, demonstrating environment-specific calibration of stickiness.

probabilistic bandit task. Although the Ideal HMM achieved marginally higher total reward (Fig. 6), it did so at the cost of substantially increased switching frequency, reflecting heightened sensitivity to stochastic reward omissions. In contrast, dynamic RFLR achieved comparable reward while maintaining lower switching rates and improved reward-per-switch efficiency. This pattern suggests that adaptive stickiness does not maximise reward in a strictly normative sense, but instead optimises a trade-off between reward acquisition and switching cost. In environments with stochastic feedback, excessive reactivity can degrade behavioural efficiency even if it remains inference-consistent. From this perspective, stickiness can be interpreted as a bounded-rational control mechanism: rather than implementing full latent-state inference, agents may regulate behavioural persistence to stabilise decisions under uncertainty. In this sense, stickiness may serve as a low-dimensional control signal that approximates belief updating without explicitly representing latent states. Adaptive modulation of stickiness allows persistence to scale with environmental volatility, increasing when feedback becomes unreliable and thereby preventing over-switching. Thus, dynamic stickiness may approximate the efficiency trade-offs associated with inference-based behaviour in a resource-efficient manner, sacrificing minimal reward while substantially reducing switching cost. This interpretation reframes stickiness not as a suboptimal bias, but as an efficiency-oriented computational strategy for balancing flexibility and stability in uncertain environments.

DISCUSSION

The central contribution of this work is to demonstrate that stickiness is not merely a descriptive bias parameter, but functions as a dynamically regulated control variable that shapes behavioural efficiency and predictive performance under uncertainty.

Reproduction and validation of stochastic switching

We first successfully reproduced the principal behavioural and modelling findings of Beron et al. (2022). Consistent with the original study, mouse behaviour exhibited graded, stochastic switching that depended systematically on recent reward history. Among the tested models, the RFLR framework most accurately captured both the conditional switching structure and reversal-aligned adaptation dynamics, whereas inference-based HMM variants exhibited excessive switching due to sensitivity to stochastic reward omissions. This reproduction establishes a validated computational baseline and confirms that stickiness-based models provide a more faithful account of observed policy structure than inference-only alternatives.

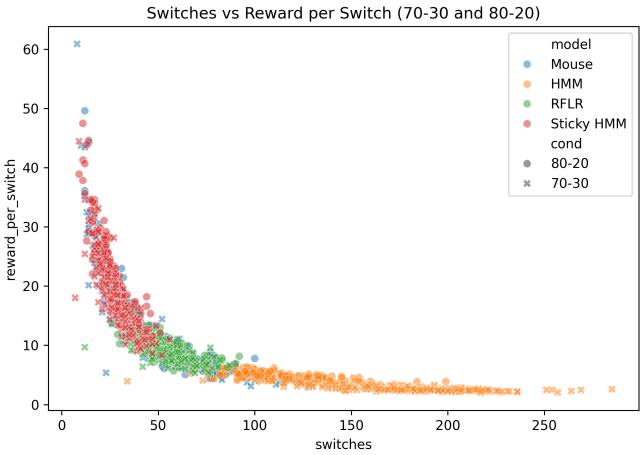


FIG. 8: **Efficiency frontier reveals stabilising role of stickiness.** Reward-per-switch efficiency plotted against switching frequency. Adaptive stickiness mechanisms shift behaviour toward lower-switch, higher-efficiency regimes, reducing unnecessary switching while maintaining reward.

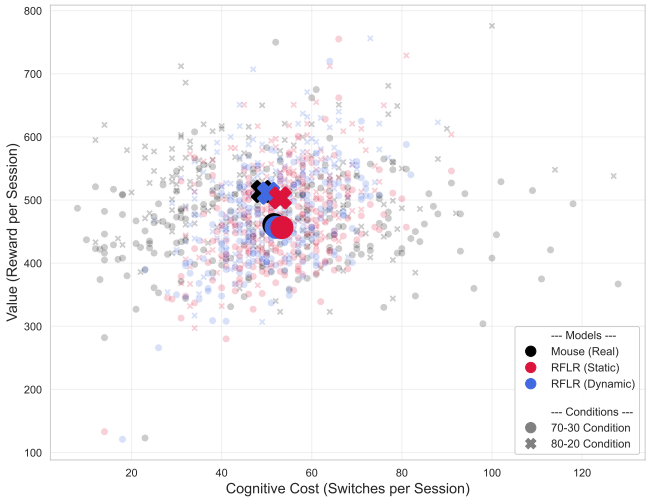


FIG. 9: **Dynamic stickiness improves efficiency relative to static RFLR.** Comparison of static and dynamic RFLR models across reward environments. Dynamic stickiness reduces switching frequency while preserving reward, shifting behaviour toward the empirical mouse regime, particularly under increased uncertainty.

Stickiness increases with environmental uncertainty

Extending the task to the noisier 70–30 environment revealed systematic modulation of fitted parameters. Cross-validated estimates (Table I) showed that base stickiness increased substantially under increased reward stochasticity, whereas surprise sensitivity decreased, indicating reduced trial-by-trial reactivity in noisier environments. Importantly, these shifts were accompanied by improvements in held-out predictive log-likelihood for dynamic RFLR relative to alternative adaptive variants across both environments. Although adaptive variants introduce additional parameters (e.g., γ and commitment window size N), improvements in held-out likelihood indicate that gains reflect improved predictive structure rather than mere parameter flexibility. Thus, increased stickiness under uncertainty reflects not only descriptive efficiency but also stable generalisation to unseen data.

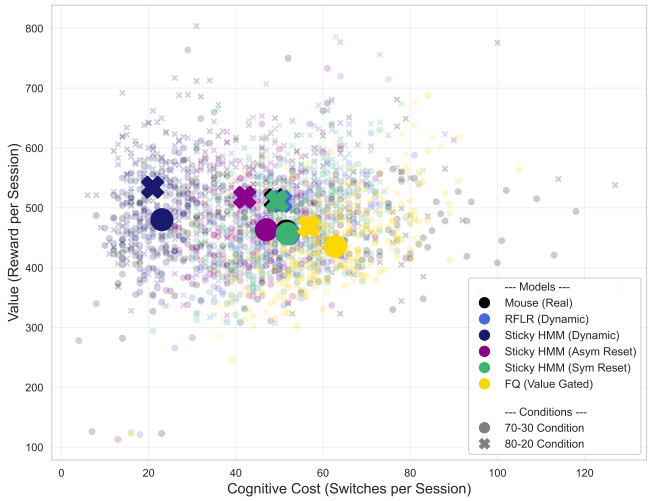
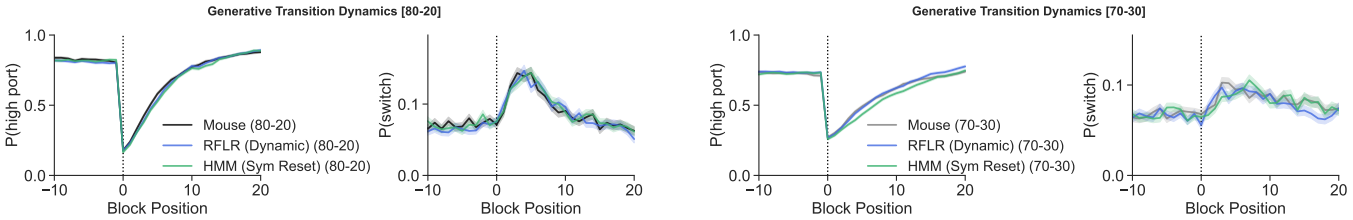


FIG. 10: **Adaptive models shift behaviour toward the mouse regime.** Efficiency frontier comparison across adaptive variants. Dynamic RFLR, Sticky HMM reset models, and value-gated F-Q occupy regions of higher reward-per-switch efficiency than inference-only baselines, demonstrating improved stabilisation under stochastic feedback.

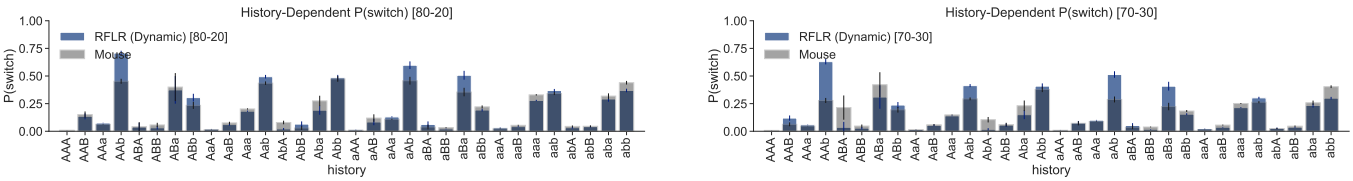
Adaptive stickiness improves efficiency and predictive performance

Generative analyses demonstrated that adaptive modulation of stickiness improves reward-per-switch efficiency by reducing excessive switching while preserving reward acquisition. Crucially, likelihood-based comparison confirmed that these adaptive mechanisms enhance predictive fidelity to observed choice sequences. Across both environments, dynamic RFLR achieved the highest held-out log-likelihood among adaptive variants, and fold-wise performance suggested stable generalisation rather than fold-specific effects. Together, the generative and predictive results converge on the interpretation that adaptive stickiness stabilises behaviour in noisy environments without sacrificing flexibility.



(a) 80–20 (b) 70–30

FIG. 11: **Adaptive stickiness stabilises reversal-aligned dynamics.** Generative transition dynamics aligned to contingency reversals for both reward environments. Under increased uncertainty (70–30), adaptive stickiness produces more gradual and bounded switching, preventing oversensitive responses to stochastic losses.



(a) 80–20 (b) 70–30

FIG. 12: **Dynamic RFLR reproduces graded history-dependent switching.** Conditional switching probabilities across all 32 three-trial reward histories. Dynamic stickiness preserves bounded, graded switching under both reward environments, closely resembling empirical mouse policy structure.

Stickiness as a computational control variable

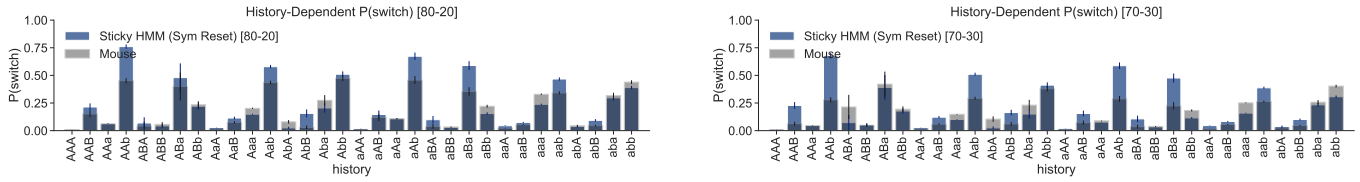
These findings support a reinterpretation of stickiness as a functional control parameter regulating the flexibility–stability trade-off. Fixed stickiness improves behavioural stability but cannot adapt across environments. Dynamic stickiness, by contrast, allows persistence to scale with environmental volatility, increasing when stochasticity is high and decreasing when feedback is reliable. From this perspective, stickiness can be understood as a bounded-rational regulatory mechanism: rather than implementing full latent-state inference, agents may approximate the efficiency trade-offs associated with inference-based behaviour through low-dimensional modulation of behavioural persistence. This control-oriented interpretation situates stickiness within a broader computational framework in which decision systems regulate gain or persistence in response to environmental volatility. Rather than representing latent states explicitly, adaptive persistence may serve as a computationally efficient surrogate for belief updating, capturing key behavioural signatures while reducing sensitivity to stochastic noise. In this sense, dynamic stickiness provides a parsimonious mechanism for balancing flexibility and stability in sequential decision problems.

Limitations and future directions

Several limitations remain. First, although adaptive variants were evaluated using cross-validated likelihood, more comprehensive hierarchical fitting across animals would strengthen statistical inference and permit formal model comparison at the population level. Second, simulations were conducted over a limited number of synthetic sessions, which may constrain statistical power for generative comparisons. Third, the present framework remains algorithmic rather than mechanistic; future work should examine whether adaptive stickiness can be embedded within biologically plausible circuit models and whether neural correlates of persistence modulation are observable in relevant brain regions. Overall, the present findings demonstrate that dynamic regulation of behavioural persistence constitutes a robust and predictive computational strategy for decision-making under uncertainty. By integrating reproduction, cross-environment analysis, and adaptive modelling, this work advances a principled account of stochastic switching as an efficiency-oriented, dynamically regulated process.

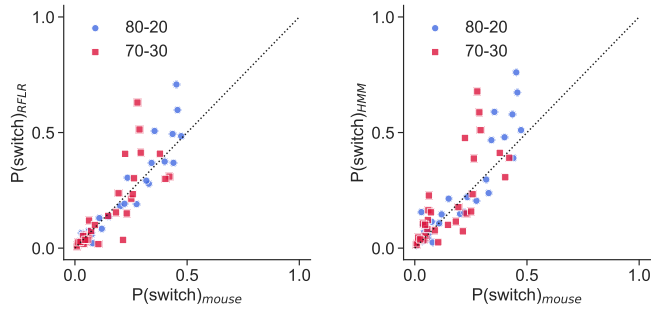
CONCLUSION

This study reproduced and extended prior work on stochastic switching in probabilistic bandit tasks and demonstrated that stickiness plays a fundamental computational role in adaptive decision-making. First, we



(a) 80–20 (b) 70–30

FIG. 13: **Sticky HMM with symmetric reset improves graded policy structure.** History-dependent switching probabilities for the symmetric-reset HMM variant. Reset mechanisms reduce excessive switching relative to inference-only HMM, though switching remains slightly elevated following rare loss streaks.



(a) RFLR (Dynamic) (b) Sticky HMM (Sym Reset)

FIG. 14: **Adaptive stickiness improves quantitative policy agreement.** Correlation between model-predicted and empirical conditional switching probabilities. Dynamic RFLR exhibits strong alignment across environments, while Sticky HMM reset variants show moderate but improved agreement relative to inference-only models.

successfully replicated the graded, history-dependent switching structure reported by Beron et al. (2022), confirming that stochastic behavioural persistence is more accurately captured by stickiness-based models than by inference-only HMM frameworks. Second, extending the task to a noisier 70–30 environment revealed systematic modulation of fitted parameters, with increased base stickiness and reduced surprise sensitivity under greater reward uncertainty. These shifts were supported by cross-validated improvements in predictive log-likelihood, indicating that behavioural persistence adapts in a cross-validated and generalisable manner. Third, we showed that adaptive, surprise-dependent modulation of stickiness improves behavioural efficiency and predictive performance. Across reward environments, dynamic RFLR achieved the highest held-out likelihood and occupied a favourable position on the reward–switch efficiency frontier, reducing excessive switching while preserving reward acquisition. These findings suggest that stickiness is not merely a descriptive bias, but functions as a dynamically regulated control parameter governing the flexibility–stability trade-off. More broadly, the results support an interpretation of stickiness as a bounded-rational regulatory mechanism. Rather than implementing full latent-state inference, agents may approximate the efficiency trade-offs associated with inference-based behaviour through low-dimensional modulation of behavioural persistence. Adaptive regulation of stickiness allows behaviour to scale with environmental volatility, providing a computationally efficient strategy for stabilising decisions under stochastic feedback. Together, these findings highlight dynamic behavioural persistence as a general computational principle for balancing flexibility and stability in uncertain environments, offering a parsimonious and predictive account of stochastic switching behaviour.

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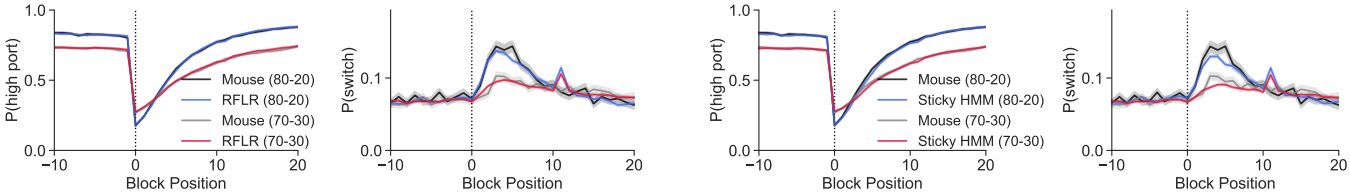
* All figures were generated using custom Python code written by the authors, unless stated otherwise, e.g., for reproduction of original results. The full analysis and figure-generation pipeline is available at `https://github.com/BOBRICE1/Stochastic-Switching-Bandits`.

[1] C. C. Beron, J. Nehoran, A. J. Moss, M. R. Veit, and M. R. Cohen, Proc. Natl. Acad. Sci. U.S.A. **119**, e2113961119 (2022).

[2] A. Izquierdo, J. L. Brigman, A. K. Radke, P. H. Rudebeck, and A. Holmes, Nat. Rev. Neurosci. **18**, 655 (2017).

[3] N. D. Daw, Y. Niv, and P. Dayan, Nat. Neurosci. **8**, 1704 (2005).

[4] R. C. Wilson and Y. Niv, PLoS Comput. Biol. **10**, e1003670 (2014).



(a) Dynamic RFLR

(b) Sticky HMM (Sym Reset)

FIG. 15: **Predictive reversal dynamics confirm stabilising effect of adaptive stickiness.** Model-predicted choice and switching probabilities aligned to reversals across reward environments. Adaptive stickiness maintains bounded switching and avoids oversensitivity under increased uncertainty, closely tracking empirical mouse adaptation.